

Roadmap

1

- Stage 1 — Day 1: Introduction to Python Development**
- What is CLI (Command Line Interface)
 - Difference between Python 2 and Python 3
 - Introduction to Django
 - PEP 8 Python style guide
 - GIGO (Garbage In, Garbage Out)
 - Introduction to Anaconda

2A

- Stage 2 — Day 2: CLI Fundamentals**
- Detailed introduction to CLI
 - What is CLI and why developers use it
 - Common CLI operations
 - Useful CLI tips
 - Relative paths vs absolute paths
 - Running Python from the terminal

2B

- Stage 3 — Day 2: Git Fundamentals**
- What is Git and why it is used
 - Basic Git operations
 - Initialize a Git repository
 - Check file status
 - Add and commit changes
 - View commit history
 - Push and pull project changes

3A

- Stage 4 — Day 3: PowerShell Automation**
- Introduction to automation
 - PowerShell scripts with the .ps1 extension
 - Run .ps1 files from the terminal
 - Use Task Scheduler
 - Create automatic backups
 - Use Write-Host to display messages
 - Create folders and files automatically

3B

- Stage 5 — Day 3: Python Project Setup and Git Branching**
- Automate Python project setup using a .ps1 script
 - Create and activate a Python virtual environment
 - Install required packages
 - Save dependencies using pip freeze
 - Initialize Git automatically
 - Basic Git branching
 - Merge branches into the main branch

4

- Stage 6 — GitHub and Collaboration**
- Introduction to GitHub
 - Connect a local Git repository to GitHub
 - Configure Git username and email
 - Create and manage remote repositories
 - Push local commits to GitHub
 - Pull changes from GitHub
 - Create and switch between branches
 - Merge branches
 - Fork an existing repository
 - Contribute to projects using the Fork → Clone → Edit → Push → Pull Request workflow